Re-analysis Summary Report for Peer-evaluation

Paper ID: **Bartels_JournConsRes_2015_mrZ**

Re-analyst's ID: **50pce**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
Cues that highlight opportunity costs reduce spending primarily when people …are more connected to their future selves. (p. 1469.)

The corresponding article

Here you can find the original article: https://osf.io/rnkgf/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/h6m5d/?view_only=b23bbc54fbfb438595d8db9ac8f62ea4

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
Opportunity cost condition was dummy coded (0=control, 1 = opportunity cost reminder), and purchase intentions was recoded (0=chose not to buy, 1 = chose to buy). To test the interaction between opportunity cost condition and future connectedness, I calculated the product of these variables. In a logistic regression, I regressed purchase intentions on opportunity cost condition, future connectedness, and the product term representing their interaction. This analytical approach was used because it allows for the examination of statistical interactions between continuous and categorical predictors of a binary criterion variable, which were the features of the current data and research hypothesis. An interaction test provides a direct test of whether the effect of a predictor (i.e., opportunity cost condition) significantly varies as a function of a moderator (i.e., future connectedness). The interaction was statistically significant, B = -1.05, SE = .50, p = .036. The conditional effects of the opportunity cost manipulation were examined at low (1 SD below the mean) and high (1 SD above the mean) levels of future connectedness. To accomplish this, future connectedness was centered so zero was 1 SD below or above the mean, the product term was recomputed using one of these centered versions of future connectedness, and the regression analysis described above was conducted again after replacing the original future connectedness variable and product term with one of these centered

versions. Consistent with the prediction, the opportunity cost reminder manipulation did not significantly affect purchase intentions for those low in future connectedness, B = .24, Exp(B) = 1.27, SE = .70, p = .73; but it decreased purchase intentions for those high in future connectedness, B = -1.86, Exp(B) = .16, SE = .68, p = .006.

The analyst's conclusion: Consistent with the hypothesis, reminders of opportunity costs reduced purchase intentions only for participants high in future connectedness.

The analyst's categorization of the result:
The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should use the dataset for Study 1a for your analysis. You should treat psychological connectedness as a continuous measure.

Here, the analyst reported the following steps and results of the analysis:
Experimental condition was dummy coded (0=control condition and 1 = opportunity cost reminder condition), as was decisions to buy the DVD (0=decided not to buy; 1 = decided to buy). I conducted a logistic regression analysis in which purchase decisions was regressed on experimental condition, connectedness, and their product, representing their interaction. The hypothesis tested was that cues that highlight opportunity costs reduce spending (relative to the control condition) more when participants score high, relative to low, in future connectedness. Support for this hypothesis would be represented as a significant and negative interaction effect. I chose to use logistic regression because the criterion variable is binary, and because this statistical approach provides a single statistical result (the test of the interaction) that would serve as a suitable test of the hypothesis, which was required for task 2. Consistent with the hypothesis, the interaction was significant and negative, B = -1.05, SE = .50, Wald = 4.41, p = .036, Exp(B) = .35.

Analysis language/software/code (optional to check):
Task 1: SPSS
Task 2: SPSS
Analysis codes: https://osf.io/xsbyz/